

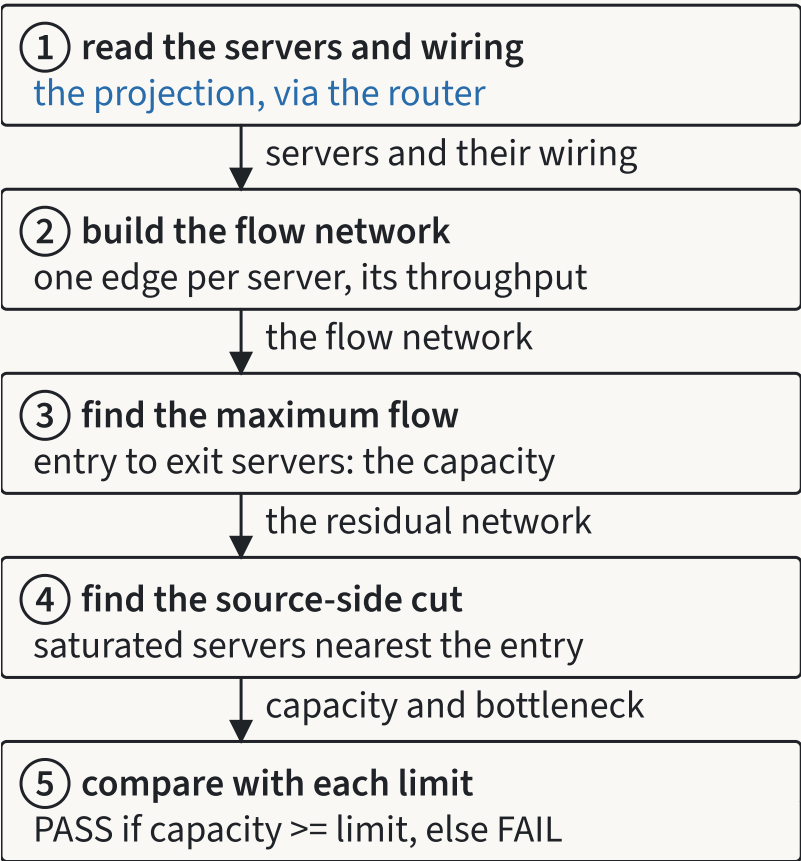
L2b Pipeline example: capacity and verdicts

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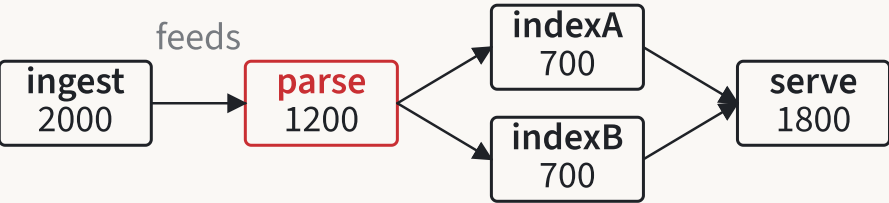
Why does raising one server change nothing and raising another change everything?
the capacity service and the example model, after the demo has been run once

functional view: how the number is made

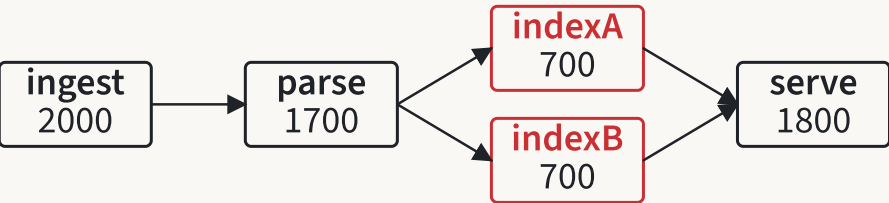


known issue: limits of validity
exact for the idealised model,
an upper bound for a real one that
meets the assumptions, and not for
capacity planning. The assumptions:
flow is conserved at every server
even partitioning across branches
perfect load balancing
no queueing, no latency coupling
stationary load, no bursts
unlimited connections

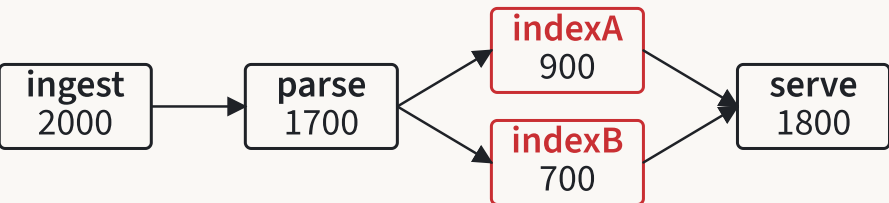
the wiring, in queries per second
shipped: 1200, PIPE-R1 **FAIL**



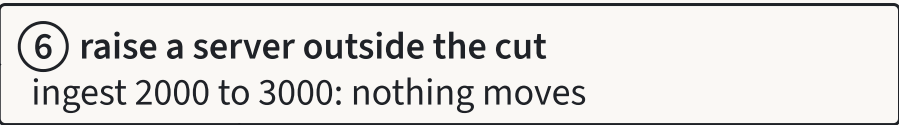
parse to 1700: 1400, PIPE-R1 **FAIL**



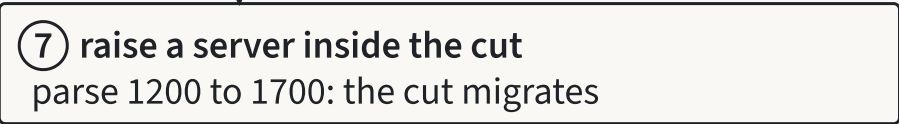
then indexA to 900: 1600, PIPE-R1 **PASS**



verdicts and reasons



the same number, the same cut



then indexA 700 to 900: capacity 1600,
PIPE-R1 **PASS**, and PIPE-R1.4 on indexB
still fails, 700 against its allocated 750

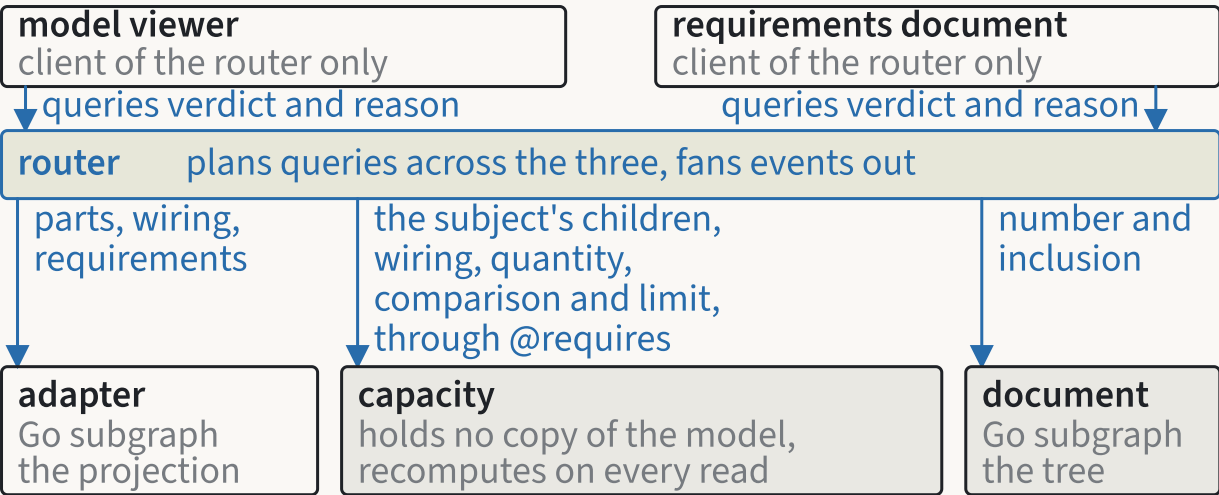
the arithmetic

capacity = the maximum flow, entry to exit servers, each
server one edge sized by its throughput = the minimum cut:
min over a chain, sum over parallel branches

state	cut	capacity, PIPE-R1	derived
shipped	parse	1200 FAIL	PASS FAIL FAIL FAIL PASS
ingest 3000	parse	1200 FAIL	PASS FAIL FAIL FAIL PASS
parse 1700	indexA indexB	1400 FAIL	PASS PASS FAIL FAIL PASS
indexA 900	indexA indexB	1600 PASS	PASS PASS PASS PASS FAIL PASS

derived, in order: ingest, parse, indexA, indexB, serve
limits: 1500 on the pipeline, ingest, parse and serve,
750 on each index server. Row 2 is box 6, row 4 is box 7:
PIPE-R1.4 fails at 700 against its allocated 750 while
PIPE-R1 passes
reasons capacity 1200 against 1500, limited by parse
capacity 1400 against 1500, limited by indexA, indexB
capacity 1600 against 1500, limited by indexA, indexB
all values known, from the running example,
guarded by a unit test

where it runs: three services, one router



decisions

- ★ AD-0006 idealised model
only throughput limits a server

★ AD-0024 reasons from fixed templates
a reason carries no word of the model
the record: docs/architecture.md and docs/capacity-model.md.
Sibling sheets: L0, L1, L2a. Decision records: AD-0006, AD-0007, AD-0024, AD-0025.
- ★ AD-0007 maximum flow, source-side cut
any wiring, the nearest cut is unique

★ AD-0025 the document owns its structure
it owns order, nesting and numbering

- a component

■ owned by the example

□ the router and what
passes through it

□ a failing verdict or
the bottleneck
- ★ a decision record, its id
and one line of rationale
(the cookbook types stars
by concern)

① the reading path
1200 a known value